Justin Manoj

845-633-1106 justinmmanoj@gmail.com linkedin.com/justinmmanoj github.com/justinm5

Education

University of Massachusetts Amherst Expected May 2026
Bachelor of Science in Computer Science ***GPA: 3.82 / 4.00***

- **Relevant Coursework:** Data Structures and Algorithms, Computer Systems, Distributed Systems, Machine Learning, Web Programming, Computer Network and Security

Experience

IBM Fall 2026
Incoming Software Developer Co-op *Lowell, MA*

Dell Technologies May 2026 – August 2026
Software Engineer Intern *Hopkinton, MA*

- Owned end-to-end design and deployment of a Go-based Kubernetes operator that automated audit logging across production RKE2 clusters, replacing manual configuration.
- Integrated Syslog and OpenTelemetry Collector pipelines to stream compliance events into a distributed observability platform for real-time security monitoring.
- Shipped 20+ GitOps-managed releases across production clusters, including platform features, critical bug fixes, and CVE remediations.

GBCS Group March 2025 – September 2025
Software Engineer Intern *New York, NY*

- Migrated 8 REST endpoints to a GraphQL API with DataLoader batching, reducing database queries and response payloads for internal services.
- Cut high-traffic query latency by 65% using PostgreSQL composite indexes and Redis caching.
- Built a GitHub Actions CI/CD pipeline with Jest and Sentry, reducing production bug reports per sprint.

University of Massachusetts Amherst November 2024 – April 2025
Research Assistant *Amherst, MA*

- Stress-tested 7 multi-agent LLM frameworks across 10,000+ simulation runs, uncovering failure patterns and benchmarking agent reliability.
- Accelerated log analysis with Ray and Pandas, building an automated pipeline that classifies agent failures and surfaces recurring patterns.
- Defined and tracked reliability metrics for multi-agent LLM systems, producing benchmark results used to compare framework behavior at scale.

Projects

NutriPlan | *Java, Spring Boot, Python, PostgreSQL, React, Redis*

- Built a prototype AI meal planning app for UMass Dining, aggregating live menus from 4 dining halls with real-time dietary filters and LLM-generated meal suggestions.
- Designed a Spring Boot REST API with Redis caching, PostgreSQL, and a Python scraping pipeline for live menu accuracy; pending adoption into the official UMass Dining App.

PokerIQ | *Python, Go, React, PostgreSQL, Redis*

- Built a poker decision-training tool that runs Monte Carlo simulations across 30,000+ generated spots to score user actions and identify strategic leaks.
- Designed a Redis-backed adaptive quiz pipeline that tracks user error patterns and personalizes drills to maximize improvement in high-EV decision categories.

Technical Skills

Languages: Python, Java, Go, JavaScript/TypeScript, SQL
Frameworks: Spring Boot, FastAPI, React, Node.js, GraphQL, gRPC
Developer Tools: Kubernetes, Docker, Kafka, PostgreSQL, Redis, Elasticsearch, OpenTelemetry, Prometheus, Grafana, GitHub Actions, Terraform, AWS (EC2, EKS, S3)